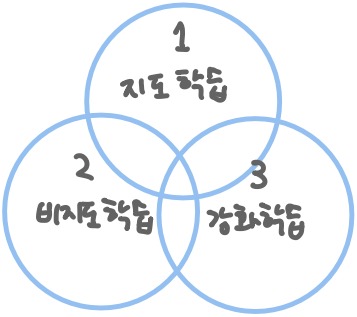
CS229. 1강

∴ 앞으로 진행할 수업들에 대한
개략적인 설명.

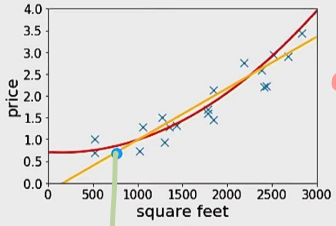


1. supervised learning

Given: a dataset that contains n samples.

$$(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)})$$

➢ Task: if a residence has x square feet, predict its price?



선형회귀분석
OR 예측함수.

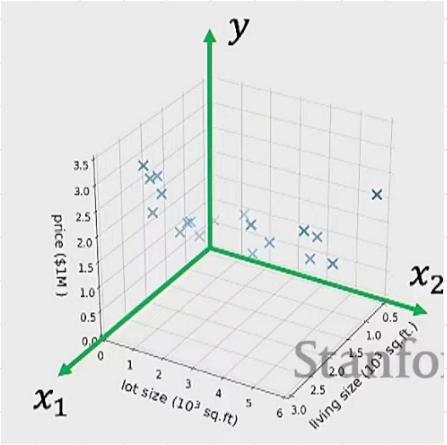
$x=800 \rightarrow y=?$

- More feature

$$\underbrace{(\text{size}, \text{lotsize})}_{\text{features / input}} \longrightarrow \underbrace{\text{price}}_{\text{label / output}}$$

$x \in \mathbb{R}^2$ $\qquad$ $y \in \mathbb{R}$

$(x_1^{(i)}, x_2^{(i)})$



• 고차원 특징들. (High-dimensional features.)

> $x \in \mathbb{R}^d$ for large d

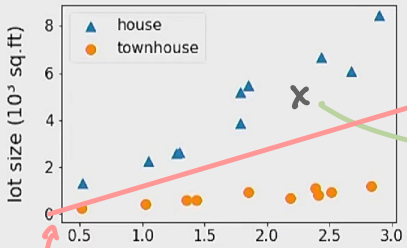
> e.g.

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_d \end{bmatrix} \begin{array}{l} \text{- living size} \\ \text{- lot size} \\ \text{- \# floors} \\ \vdots \end{array} \Longrightarrow y = \text{price.}$$

label의 종류

- 회귀 (if $y \in \mathbb{R}$ is a continuous variable) 실수값.
- 분류 (0 or 1 · 2 ··)
  (the label is a discrete variable)
  불연속적 데이터.

(size, lot size) → y: house or townhouse?

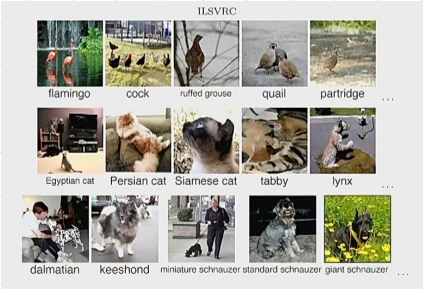


Q: house or town house?

선형 분류기.

- Image Classifition

여러 이미지와 주어지고,
각 이미지에는 해당 이미지의 주요 객체 내용을 설명하는 label 有
(한 이미지에 하나의 중요한 물체)

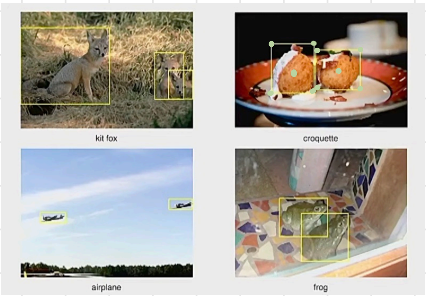


↳ (Image net DataSet)

숫자
이미지를 정렬로.)

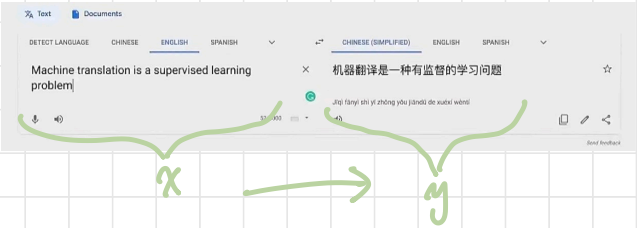
x = raw pixels of the Image (원시 픽셀값,
y = the main object.

- 객체 위치 파악 / 감지



x = raw pixels of the image
y = the bounding boxes.
(4개 숫자)

- 기계번역

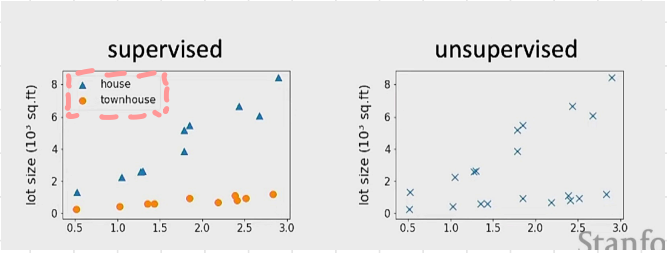


x → y

y: 분류의 문제만 다룬 것이다.
) 생각할 모든 문제의 집합. (유한하거나 무한함)

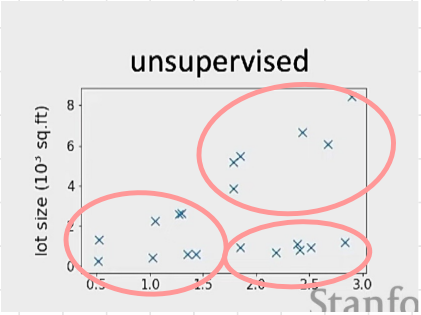
2. Unsupervised Learning

DataSet contain No Lables : $x^{(1)}, \dots, x^{(m)}$

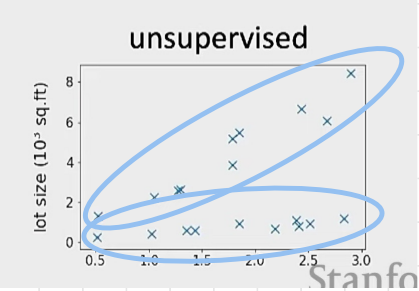


목표: 흥미로운 구조 찾기 (데이터 안에서)

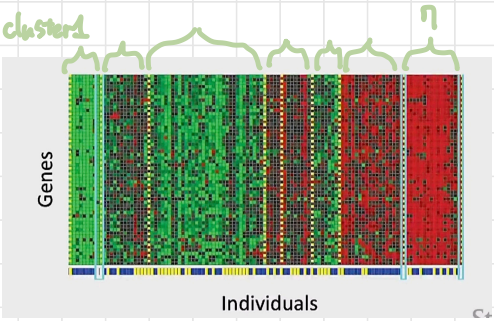
- 클러스터링.



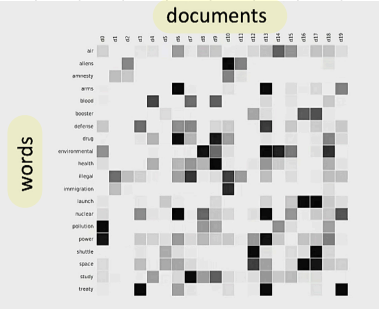
(훌륭한 알고리즘!!) → But 어쩔지 Label 이름 알수 X



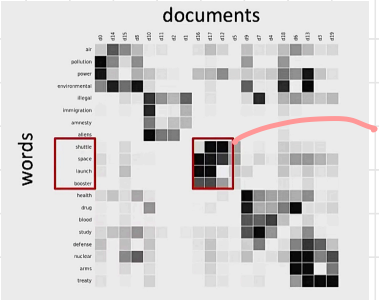
(유전자 클러스터링)



• 잠재의미분석.(LSA) : 단어의 빈도!



돌려바꿔서
대각선 중심으로..
(좋아.)

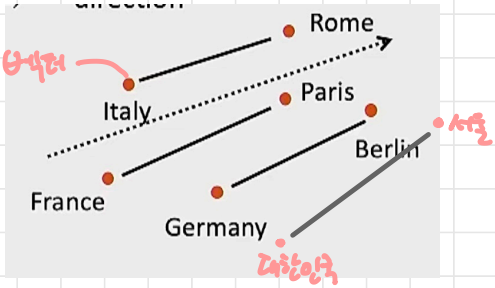


우주 관련된
문서? (추측)

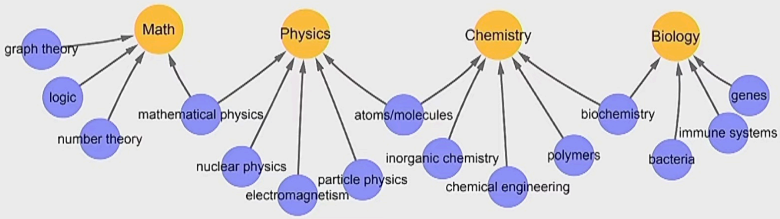
• 단어 임베딩.

word ——(encode)——→ vector

relation ——→ direction.



• 비슷한 단어 클러스터링. (벡터표현)



	logic deductive propositional semantics	graph subgraph bipartite vertex	boson massless particle higgs	polyester polypropylene resins epoxy	acids amino biosynthesis peptide
tag	*logic*	*graph theory*	*particle physics*	*polymer*	*biochemistry*

• 대규모 언어모델

- 위키피디아 같은 데이터셋 사용.

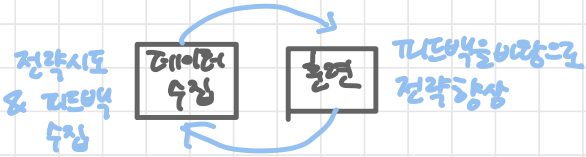
3. 강화학습.

결정과 예측의 차이?

결정에는 장기적인 결과가 따른다.

강화학습은 연속의 결정을 내려야 할 때 쓴다. (e.g. 바둑, 로봇 움직임 제어)

강화학습 알고리즘



※ But try가 영원하진 X.
경우에 따라..
e.g. 원자력발전소 제어

로봇이 넘어짐 (NewData) → 알고리즘 → 전략변경!

try ↑ ⇝ Data ↑

- Deep learning

신경망을 모델 매개변수에 사용하는 기술.

가중치 / 편향